Finite State Automata

A finite state automation can be represented by a 5 tuples, $(Q, \Sigma, \delta, q_0, F)$, when

(i) Q is a finite non-empty set of states.

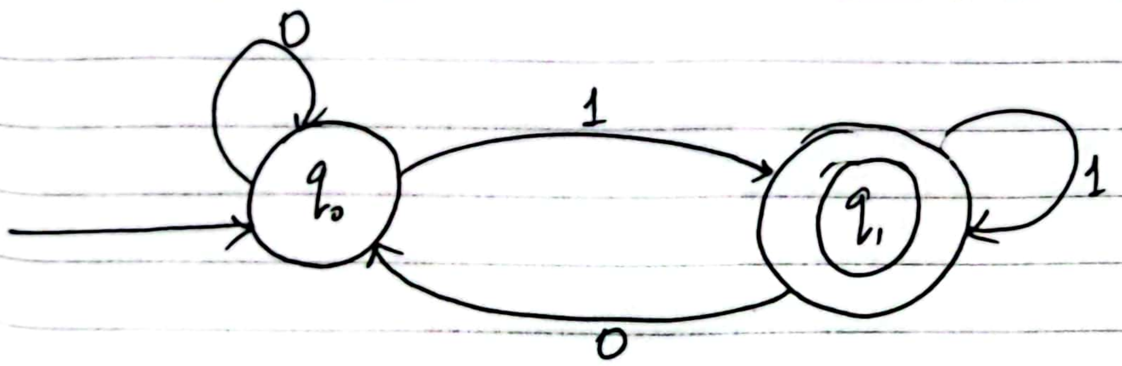
(ii) Σ is a finite non-empty set of input called input alphabet.

(iii) δ is a function which maps $Q \times \Sigma$ into Q and is called transition function.

(iv) $F \subseteq Q$ is the set of final states.

Transition Diagram

A transition diagram is a finite direct labelled graph in which each vesten represent a state and the edges are labelled as input/output.



State	Input	
	0	1
q_0	q_0	q_1
q_1	q_0	q_1

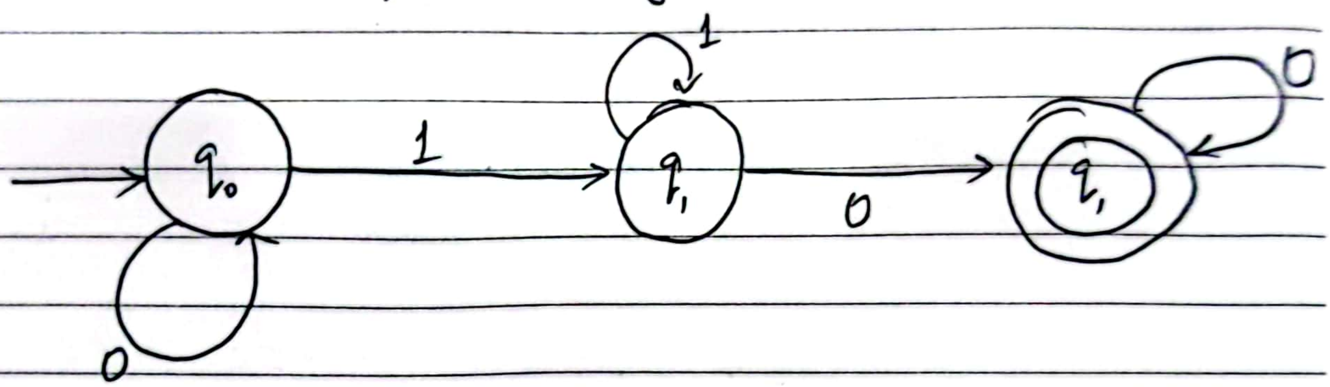
Transition Table

Notes

Types of Finite State Automation

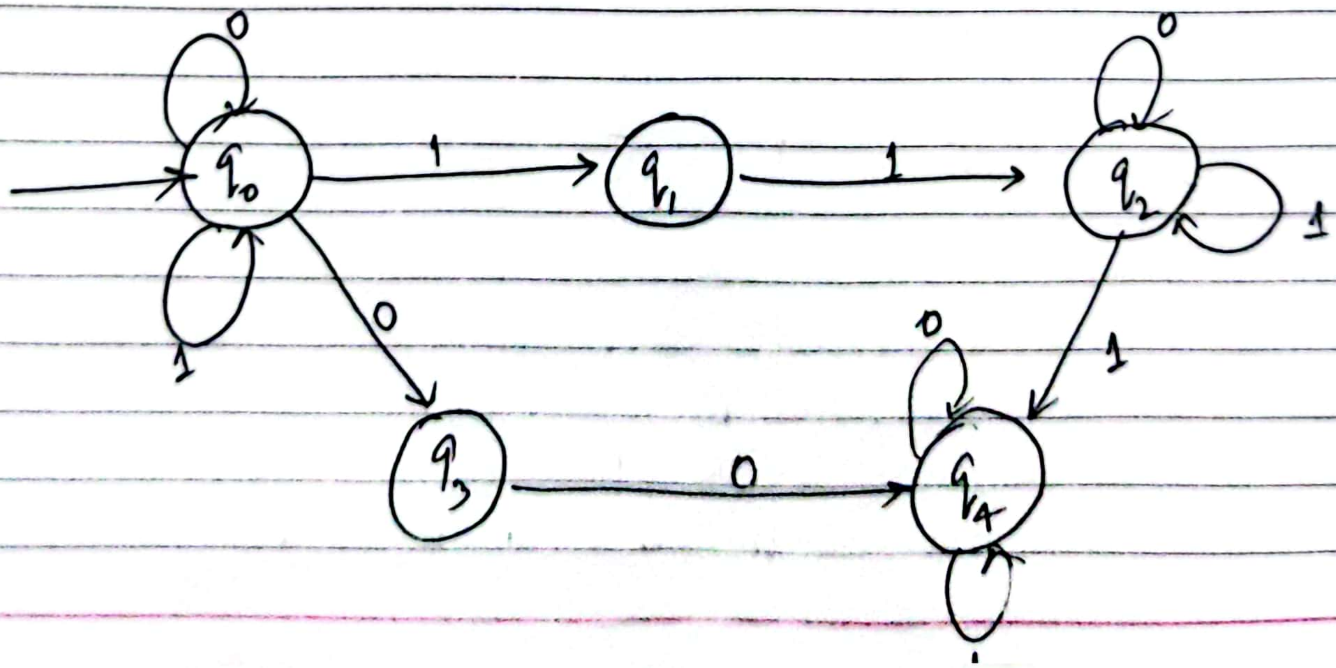
1. Deterministic Finite State Automation:

The finite automation is called deterministic finite automation if the machine is read an input string one symbol at a time



2. Non-Deterministic Finite State Automation

The finite state automation is called non-deterministic finite automation if for any particular input symbol, the machine can move to any combination of the states



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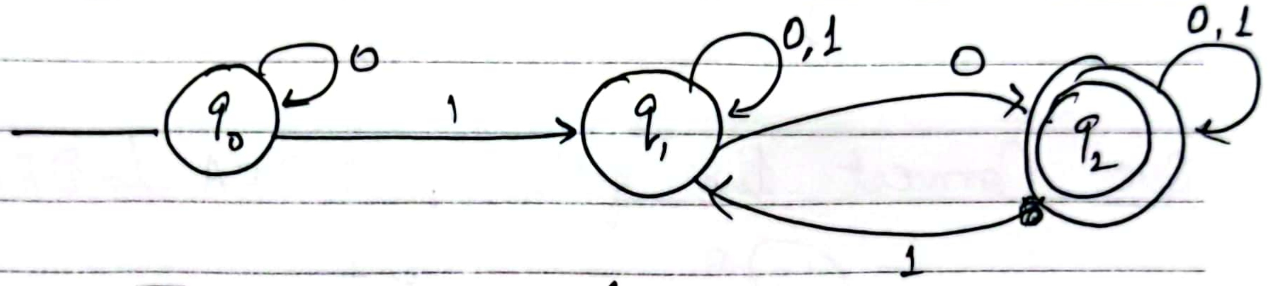
Note: Every DFA is NDFA but the converse need not to be true.

same input, different state $\Rightarrow$ NDFA

" " same state $\Rightarrow$ DFA

different " same state $\Rightarrow$ DFA

Que: Convert NDFA to DFA



Transition Table for NDFA

State	Input 0	Input 1
q_0	q_0	q_1
q_1	q_1, q_2	q_1
q_2	q_2	q_1, q_2

Sol^n: Converting into DFA

State	Input 0	Input 1
$\rightarrow q_0$	q_0	q_1
q_1	(q_1, q_2)	q_1
(q_1, q_2)	(q_1, q_2)	q_1, q_2

last row मा सबै same आएको हो भने यो DFA बन्यो।

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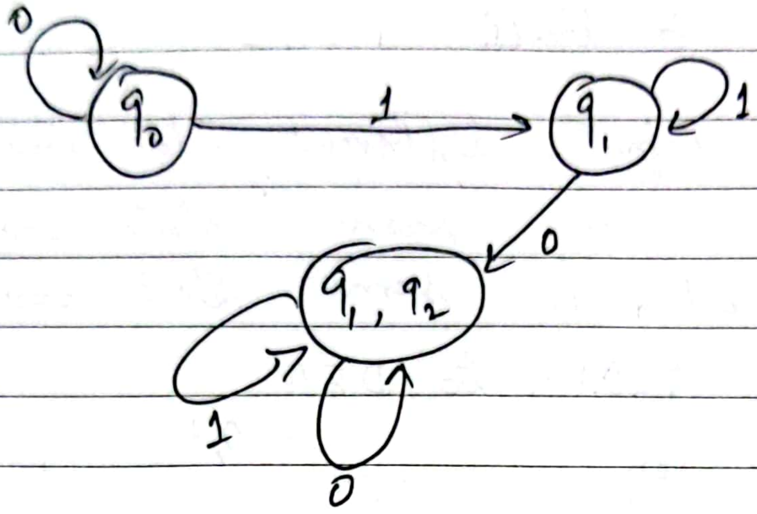
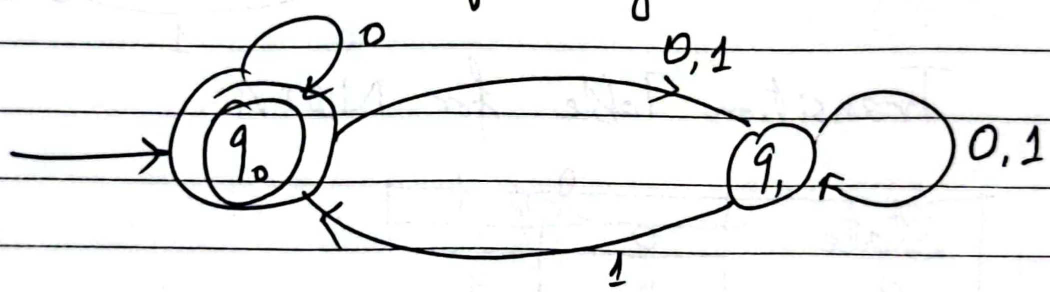
Transition Table for DFA



Que. Convert the following NDFA to DFA



Sol^n: Transition table for NDFA

State	Input	
	0	1
q_0	q_0, q_1	q_1
q_1	q_1	q_0, q_1

Converting into DFA

State	Input	
	0	1
q_0	(q_0, q_1)	q_1
(q_0, q_1)	(q_0, q_1)	(q_0, q_1)
q_1	q_1	(q_0, q_1)

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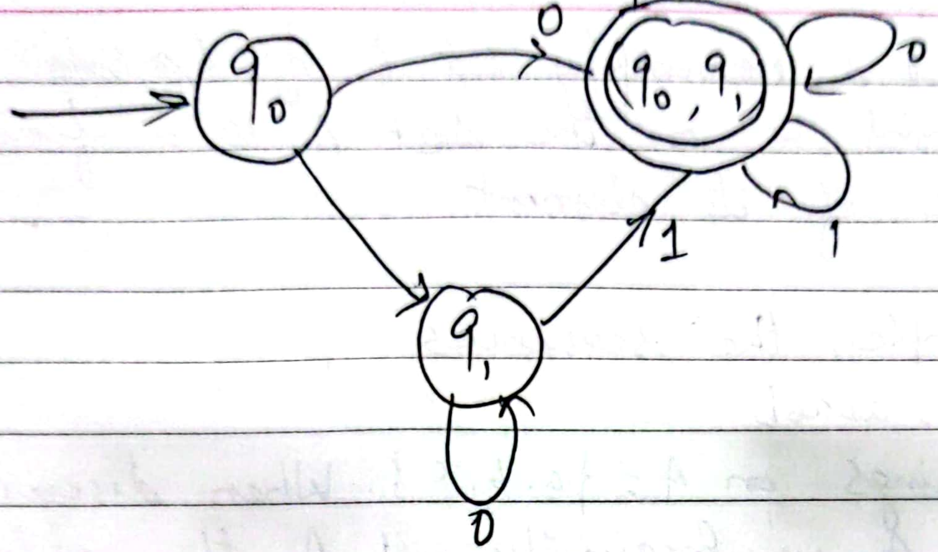
Transition Table for DFA



Notes

Strings

Consider a nonempty set A of symbols. A string/word w on the set A is a finite sequence of its elements.

For example, the sequences

$$u = ababb \quad \text{and} \quad v = accbaaa$$

are strings on $A = \{a, b, c\}$. When discussing words on A, we frequently call A the alphabet, and its elements are called letters. We will also abbreviate our notation and write a^2 for aa, a^3 for aaa and so on. Thus, for the above words,

$$u = abab^2 \text{ and } v = ac^2ba^3.$$

The empty sequence of letters, denoted by λ or ϵ is also considered to be a word on A, called the <u>empty word.</u>

The length of a word u, written $|u|$, is the number of elements in its sequence of letters.

$$\therefore |u| = 5 \text{ and } |v| = 7.$$

Also $|\lambda| = 0$, where λ is the empty word.

Concatenation

Consider two words u and v on alphabet A. The <u>concatenation</u> of u and v, written uv, is the word obtained by writing down the letters of u followed by the letters of v. For example

$$uv = ababbaccbaaa = abab^2ac^2ba^3$$

Note: $uv \neq vu$. , $\lambda u = u\lambda = u$

Reverse of String

The reverse of string u is denoted by $u^R = bbaba$ and

$v^R = aaabcca$

Prefix & Suffix: let $u = ababb$

$Pre(u) = \{\lambda, a, ab, aba, abab, ababb\}$

$Suffix(u) = \{\lambda, b, bb, abb, babb, ababb\}$

Substring let $u = ababb$

Its substrings are: a, b, a, bb, ab, aba, abab, bab, babb, abb,

Que: Find the substring of the string 'gate'.

Sol:

Substrings of length

0	1	2	3	4
λ	g	ga	gat	gate
	a	at	ate	
	t	te		
	e			

Note: If a string has distinct symbols (alphabets) then the total no. of substring that can be formed

$$= \frac{n(n+1)}{2} + 1$$

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Languages

A language L over an alphabet A is a collection of words on A. Recall that A^* denotes the set of all words on A. Thus a language L is simply a subset of A^*.

Example: Let $A = \{a, b\}$. The following are languages over A.

(a) $L_1 = \{a, ab, ab^2, \dots\}$

(b) $L_2 = \{a^m b^n : m>0, n>0\}$.

(c) $L_3 = \{a^m b^m : m>0\}$.

(d) $L_4 = \{b^m a b^n : m \geq 0, n \geq 0\}$.

Operations on languages

Suppose L and M are languages over an alphabet A. Then the "concatenation" of L and M, denoted by LM, is the language defined as follows:

$$LM = \{uv : u \in L, v \in M\}$$

That is, LM denotes the set of all words which come from the concatanation of a word from L with a word from M. e.g.

Let $L_1 = \{a, b^2\}$, $L_2 = \{a^2, ab, b^3\}$, $L_3 = \{a^2, a^4, a^6, \dots\}$ then,

$$L_1 L_2 = \{a^3, a^2b, ab^3, b^2a^2, b^2ab, b^5\}$$

$$L_1 L_3 = \{a^3, a^5, a^7, \dots, b^2a^2, b^2a^4, b^2a^6, \dots\}$$

$$L_1 L_1 = \{a^2, ab^2, b^2a, b^4\}$$

Clearly, the concatanation of languages is asso-

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Que. Write R.E. for the language accepting all strings containing any number of a's & b's.

Sol^n: R.E. = $(a+b)^*$

This will give the set as

$L = \{\lambda, a, aa, b, bb, ab, aba, ba, \ldots\}$

any combination of a & b.

The $(a+b)^*$ shows any combination with a and b, even a null string.

Que: Write the regular expression for the language accepting all the string which are starting with 1 and ending with 0, over $\Sigma = \{0, 1\}$.

Sol^n:

The first symbol is 1

last symbol is 0

$\therefore$ R.E. = $1(0+1)^*0$

Que. Write R.E. for the language starting with 'a' but not having consecutive b's.

Sol^n:

$L = \{a, aba, aab, aaa, abab, \ldots\}$

R.E. = $\{a + ab\}^*$

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Que: Write the R.E. for the language over $\Sigma = \{0\}$ having even length of the string.

$L = \{\lambda, 00, 0000, 000000, \ldots\}$

$R.E. = (00)^*$

Que: Write R.E. for the language L over $\Sigma = \{0, 1\}$ such that all the string do not contain the substring 01.

Sol^n.

$L = \{\lambda, 0, 1, 00, 11, 10, 100, \ldots\}$

$R.E. = (1^* 0^*)$